

Caleb Kang

(470) 357-1857 | cjkang2@illinois.edu | [linkedin.com/in/calebjkang](https://www.linkedin.com/in/calebjkang) | github.com/CalebK

EDUCATION

University of Illinois Urbana-Champaign

Urbana, IL

Bachelor of Science in Computer Science, Grainger College of Engineering

Aug. 2026 – May 2028

William Rainey Harper College

Palatine, IL

Associate in Engineering Science, GPA: 3.88/4.0

Aug. 2024 – May 2026

- **Awards:** Motorola Solutions Foundation Scholarship (Full-Ride), Engineering Pathways Scholarship, National Honors Society, President's List

EXPERIENCE

Software Development Engineer Intern

May 2026 – Aug. 2026

Amazon

Bellevue, WA

- Built an LLM-powered financial observability agent for Teller, Amazon's core Accounts Payable payment system (\$2.5B+ processed), translating natural-language questions to SQL over 130M+ production payment records (Python, Strands Agents SDK, Claude on Amazon Bedrock)
- Automated anomaly detection and root-cause analysis with z-score baselines and an agentic investigation loop of 20+ autonomous tool calls, delivering cited, confidence-rated explanations in 2-3 minutes; surfaced a single-day void batch of 63,469 payments (z-score 62) in production
- Deployed the containerized agent on Amazon Bedrock AgentCore Runtime with AWS CDK (TypeScript) infrastructure as code and a CI/CD pipeline; engineered least-privilege cross-account IAM access to SOX-regulated payment data via Amazon Athena, AWS Glue, Lake Formation, and S3
- Hardened the tool layer beneath the LLM with a read-only SQL validator, table allowlist, Bedrock guardrails, and anti-hallucination grounding checks; shipped 100+ unit and integration tests and validated metrics to full decimal precision across 52 month-by-business combinations

Software Engineer Intern

Jun. 2025 – Aug. 2025

Fermi National Accelerator Laboratory

Batavia, IL

- Built a fault-tolerant Python state machine to automate ASIC testing for the Deep Underground Neutrino Experiment (DUNE), supporting validation of approximately 300,000 chips across six national sites
- Mapped state transitions and failure paths in UML, implemented error handling, and defended the architecture through peer reviews and technical presentations to research scientists

AI Research Assistant

Oct. 2024 – May 2025

Biola University (Remote)

La Mirada, CA

- Fine-tuned and benchmarked a multimodal authentication pipeline using CLIP, Whisper, Sortformer, and PyTorch
- Built evaluation workflows to measure inference latency, throughput, and edge-case reliability across multimodal inputs

PROJECTS

C++ Farming Simulator | C++, OOP, Memory Management, Catch2

Oct. 2025 – Dec. 2025

- Built a terminal-based C++ simulation using inheritance, polymorphism, and a dynamic 2D pointer grid with explicit heap management, tested with a deterministic Catch2 suite covering state invariants and edge cases

Duckiebot Robotics Initiative | Linux, Docker, C++, Raspberry Pi

Dec. 2024 – Jan. 2025

- Led a 4-person team building autonomous rover controls in C++ on Raspberry Pi; planned sprint backlogs and defined deployment validation checks
- Containerized the Linux application with Docker and debugged network sockets, serial communication, and hardware configuration for reproducible deployment

TECHNICAL SKILLS

Languages: C++, Python, Java, SQL, TypeScript, JavaScript

Frameworks & Tools: AWS (Bedrock, AgentCore, Athena, Glue, Lake Formation, QuickSight, CDK), Strands Agents, PyTorch, Docker, Linux, Git, Catch2

Systems & Concepts: Data Structures & Algorithms, OOP, State Machines, Memory Management, Embedded Systems